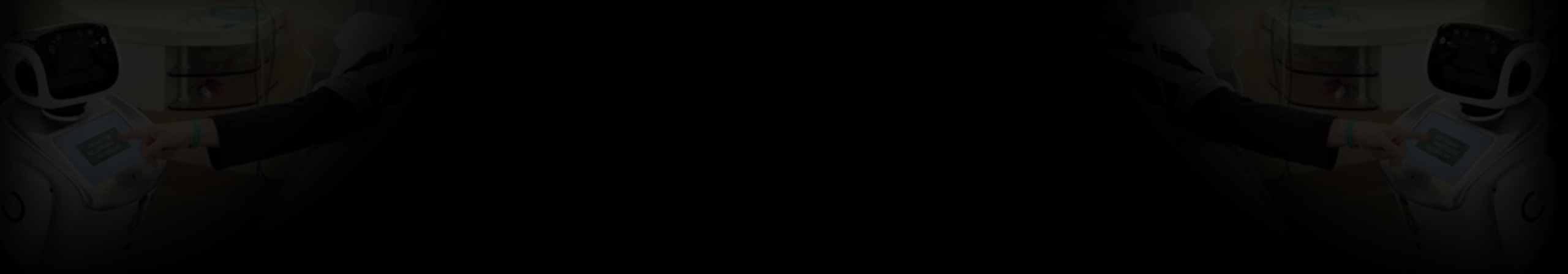


COMPUTER VISION (TFP505)

Pertemuan 7 – Feature Engineering & Image Classification

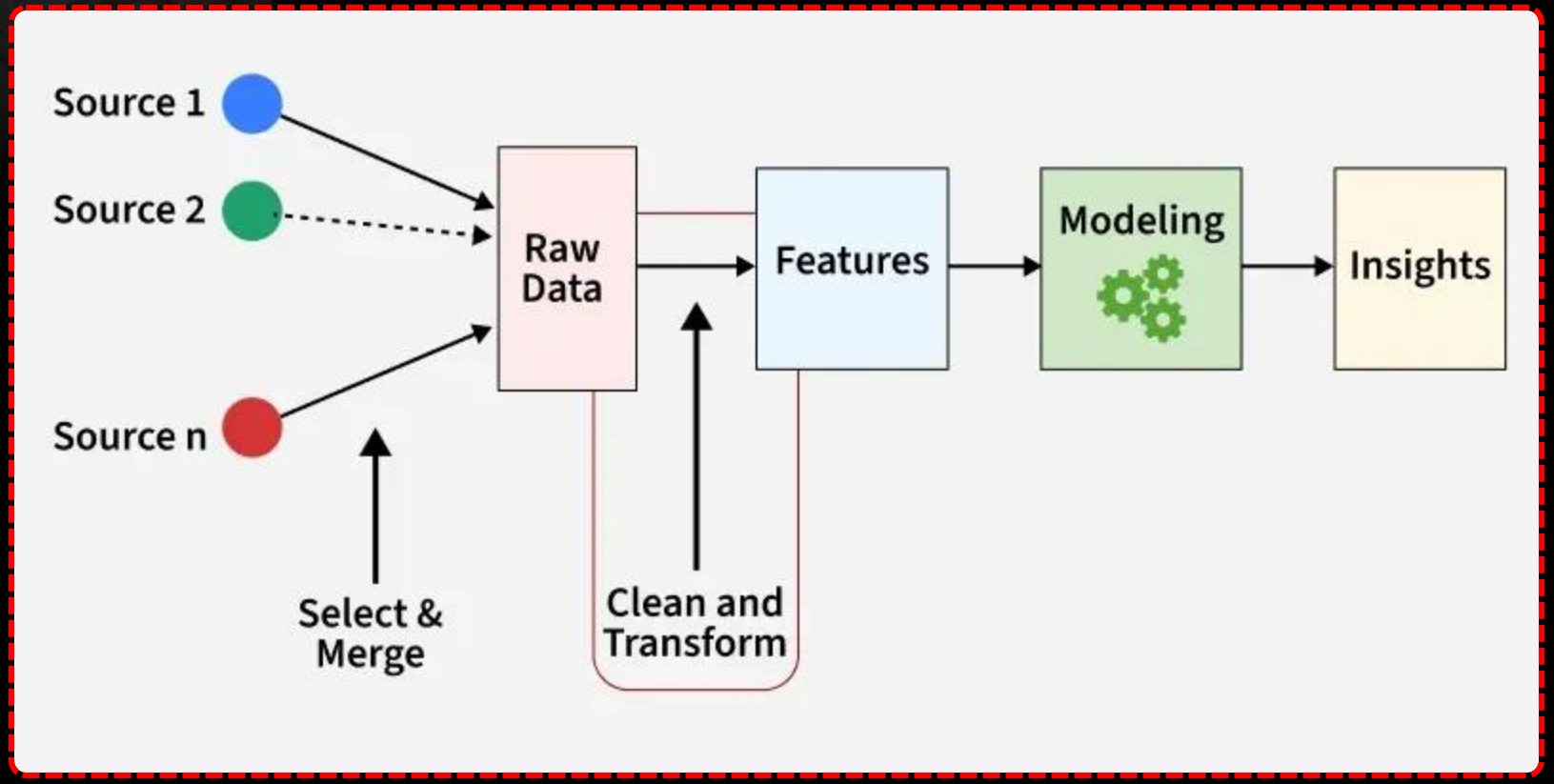
ALIFIA REVAN PRANANDA

Department of Information Technology
Faculty of Engineering
Universitas Tidar



FEATURE ENGINEERING

FEATURE ENGINEERING



- It includes choosing, creating and adjusting data attributes to make the model's predictions more accurate.
- The goal is to make the model better by providing relevant and easy-to-understand information.

FEATURE ENGINEERING is the process of turning raw data into useful features that help improve the performance of machine learning models.

FEATURE ENGINEERING

What is Feature Engineering?

A feature is a measurable property of data that is used as input for machine learning algorithms. Features can be numerical, categorical or text-based representing essential data aspects which are relevant to the problem.

Feature Creation

generating new features from observed data, e.g., we need to calculate age, but we only have the date of birth

Feature Transformation

transforming the observed data to be more suitable with the machine learning model, e.g., transforming text to numerical data

Feature Extraction

extracting information from the observed data

Feature Selection

selecting the suitable feature before employing to machine learning model

TABULAR DATASET

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22	1	0	7.25	S
1	1	1	female	38	1	0	71.28	C
2	1	3	female	26	0	0	7.92	S
3	1	1	female	35	1	0	53.10	S
4	0	3	male	35	0	0	8.05	S
...	...	...	...	...	...	...	...	...
95	0	3	male	25	0	0	8.05	S
96	0	1	male	71	0	0	34.65	S
97	1	1	male	23	0	1	63.36	S
98	1	2	female	34	0	1	23.00	S
99	0	2	male	34	1	0	26.00	S



	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	1	3	female	30	0	1	3.26	S
1	0	2	male	12	0	1	21.77	C
2	0	1	male	9	0	2	8.86	S
3	0	3	male	13	0	0	16.07	S
4	0	2	male	40	2	0	-0.09	S
...	...	...	...	...	...	...	...	...
995	0	3	female	31	0	2	40.78	S
996	0	2	female	30	1	0	12.36	S
997	1	3	female	32	1	0	-0.88	S
998	0	3	male	42	0	0	5.72	S
999	0	3	male	13	0	0	6.49	S

IMAGE DATASET

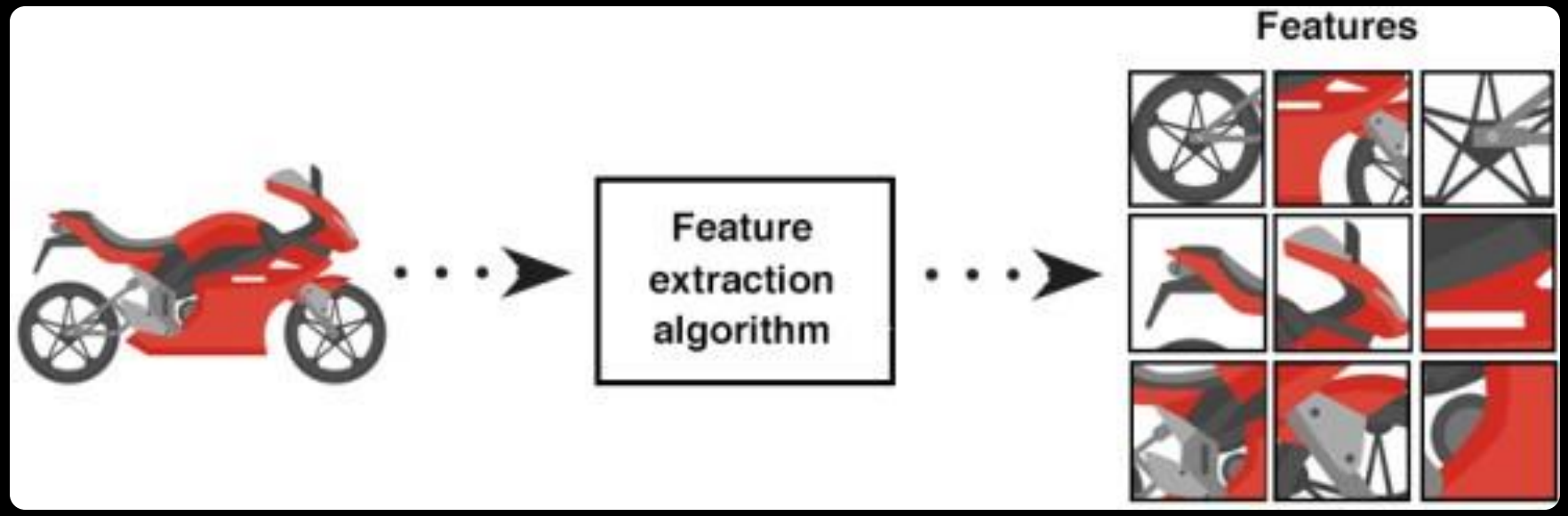


We already learn about the type of data. Feature engineering in every type of data is quite different. It depends on the need of machine learning model

FEATURE ENGINEERING

Feature Extraction

Feature extraction essentially is the process of converting raw data into numerical features that can be processed while preserving the information in the original data set. It yields better results than if you directly apply machine learning techniques on the raw data.

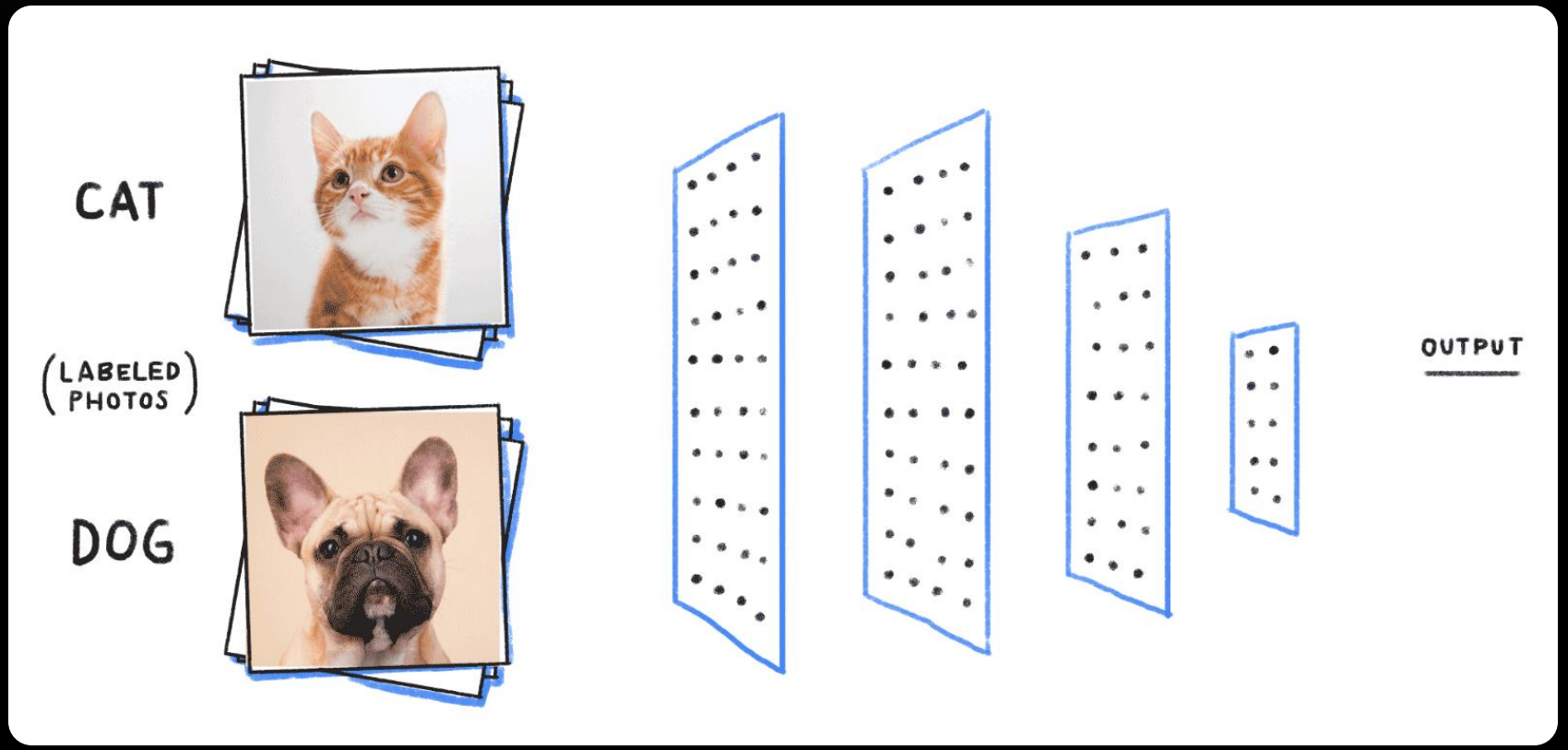


Feature defines as the main characteristic presenting the object. It can be a color, shape, or specific value such as statistical value (mean, standard deviation, etc)

If the dataset was completed with some attributes that presenting the object, hence we don't need to conduct feature extraction process.

FEATURE ENGINEERING

Feature engineering is the process of transforming raw data into meaningful features that can improve the performance of machine learning models. For images, this involves extracting important patterns such as color, texture, shape, or learned representations from deep networks. The goal is to help the model distinguish between different classes more effectively.



How can we recognize that this object is "cat" ?

What are the "main characteristics" ?

FEATURE ENGINEERING

Feature Extraction Method for Image

Feature Type	Description	Example Methods / Notes	Pros	Cons	Applications
Statistical Features	Capture statistical properties of pixel values or regions, summarizing intensity distribution and basic texture.	Mean, Standard Deviation, Skewness, Kurtosis, Color Moments	- Reduces dimensionality - Captures general intensity/color characteristics	- May lose spatial relationships - Limited for complex textures	Texture recognition, color-based classification, medical imaging
Texture Features	Represents repetitive patterns, surface structure, or spatial relationships of pixel intensities.	LBP (Local Binary Patterns), Gabor Filters, GLCM (contrast, correlation, homogeneity, energy), GLRLM (run lengths of same intensity pixels)	- Captures fine texture details - Good for material/ surface/tissue analysis	- Sensitive to rotation, scale, noise - Requires parameter tuning	Material classification, tumor texture analysis, facial expression recognition

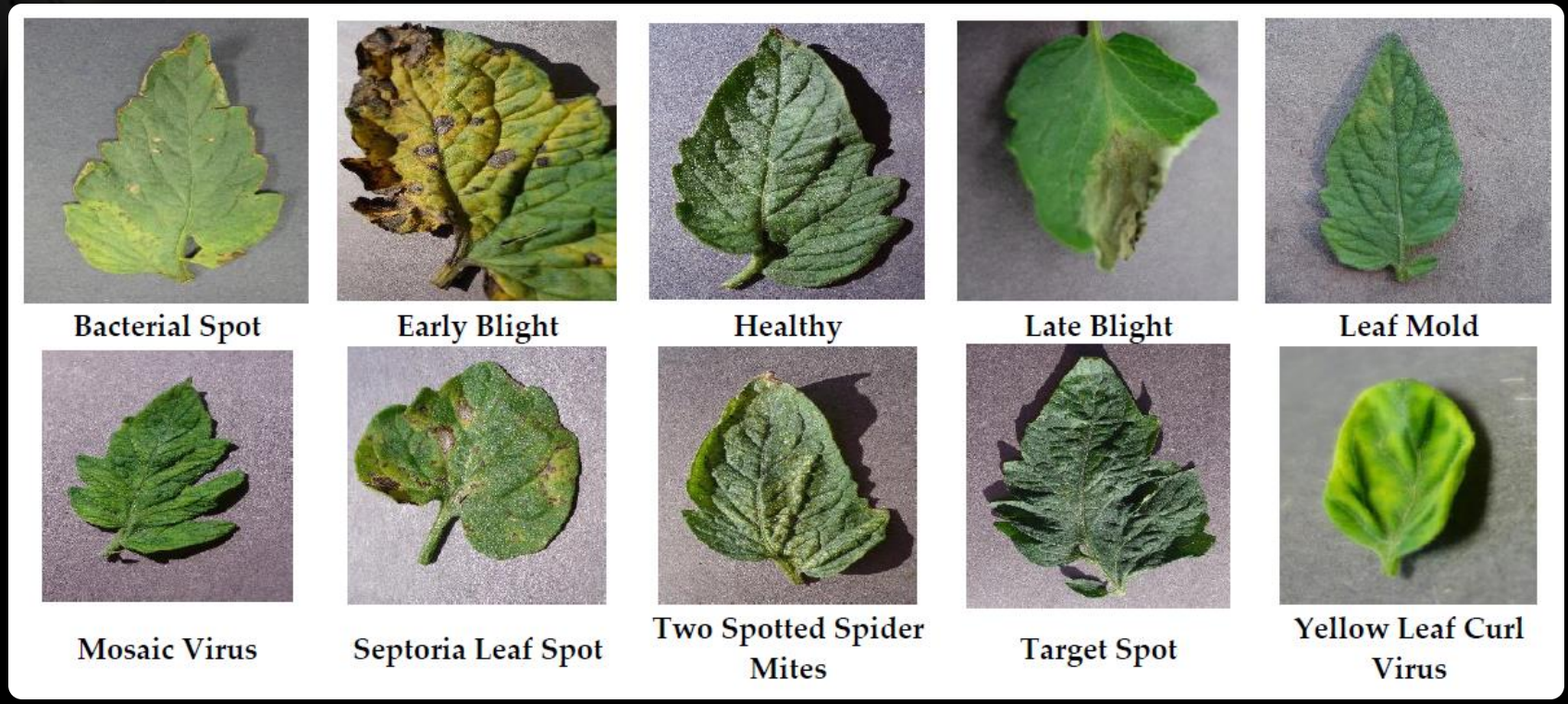
FEATURE ENGINEERING

Feature Extraction Method for Image

Feature Type	Description	Example Methods / Notes	Pros	Cons	Applications
Shape / Structural Features	Focuses on object boundaries, contours, edges, and geometric patterns.	- Contour-Based: Chain Code, Polygon Approximation, Convex Hull, Perimeter, Area - Gradient-Based: HOG, Canny Edge Detection - Moment-Based: Hu Moments, Zernike Moments - Transform-Based: Fourier Descriptors	- Captures object geometry and structure - Some methods invariant to translation, rotation, scaling	- Sensitive to noise and partial occlusion - Some methods may lose fine details	Object recognition, handwritten character recognition, traffic sign recognition, human detection

FEATURE ENGINEERING

Feature Extraction Method for Image



What are the "main characteristics" of each object ?

What are the possible "feature extraction method" that suitable with these objects ?

We can apply **"texture feature"** and **"statistical feature"** since the main differences are in the **color** and **texture**.

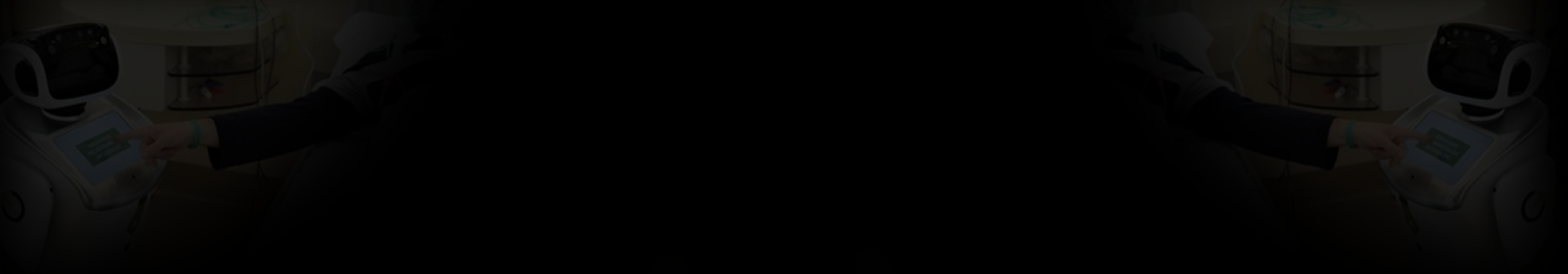


IMAGE CLASSIFICATION

IMAGE CLASSIFICATION

Image Classification: A core task in Computer Vision

Assume given set of discrete labels, e.g. {cat, dog, cow, apple, tomato, truck, … }

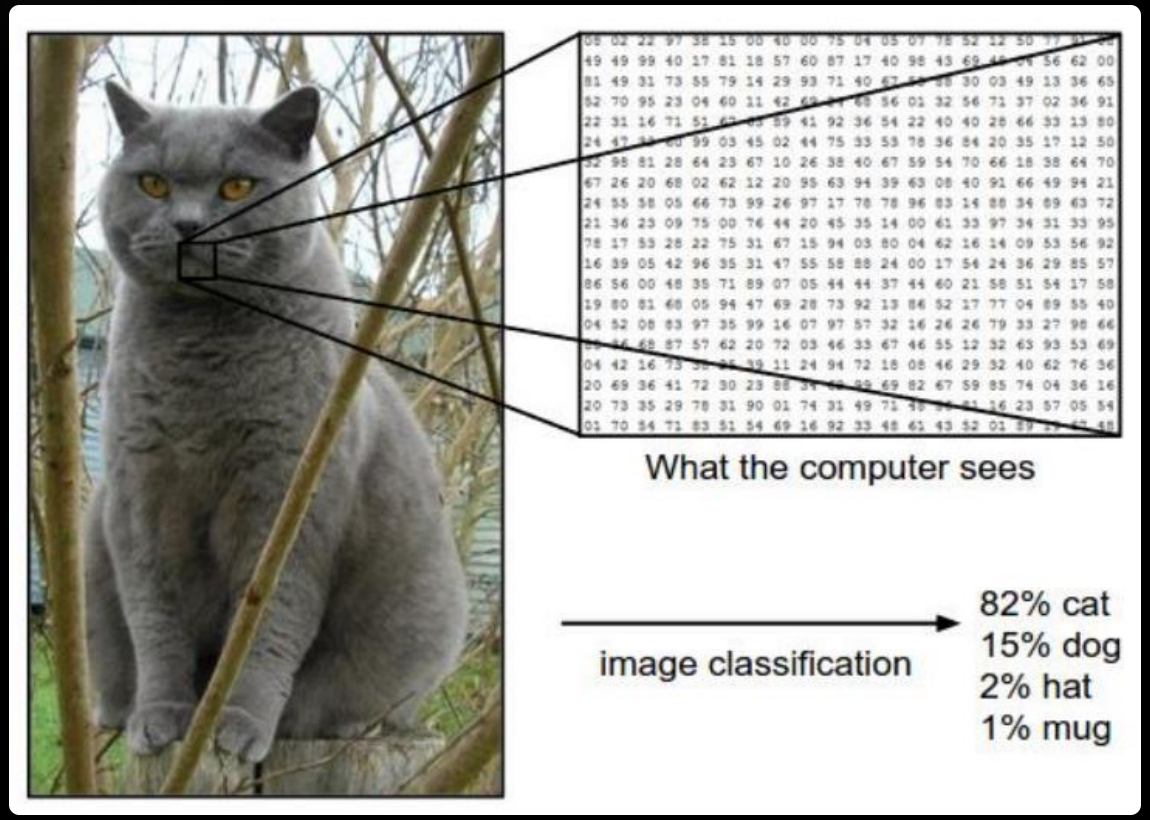
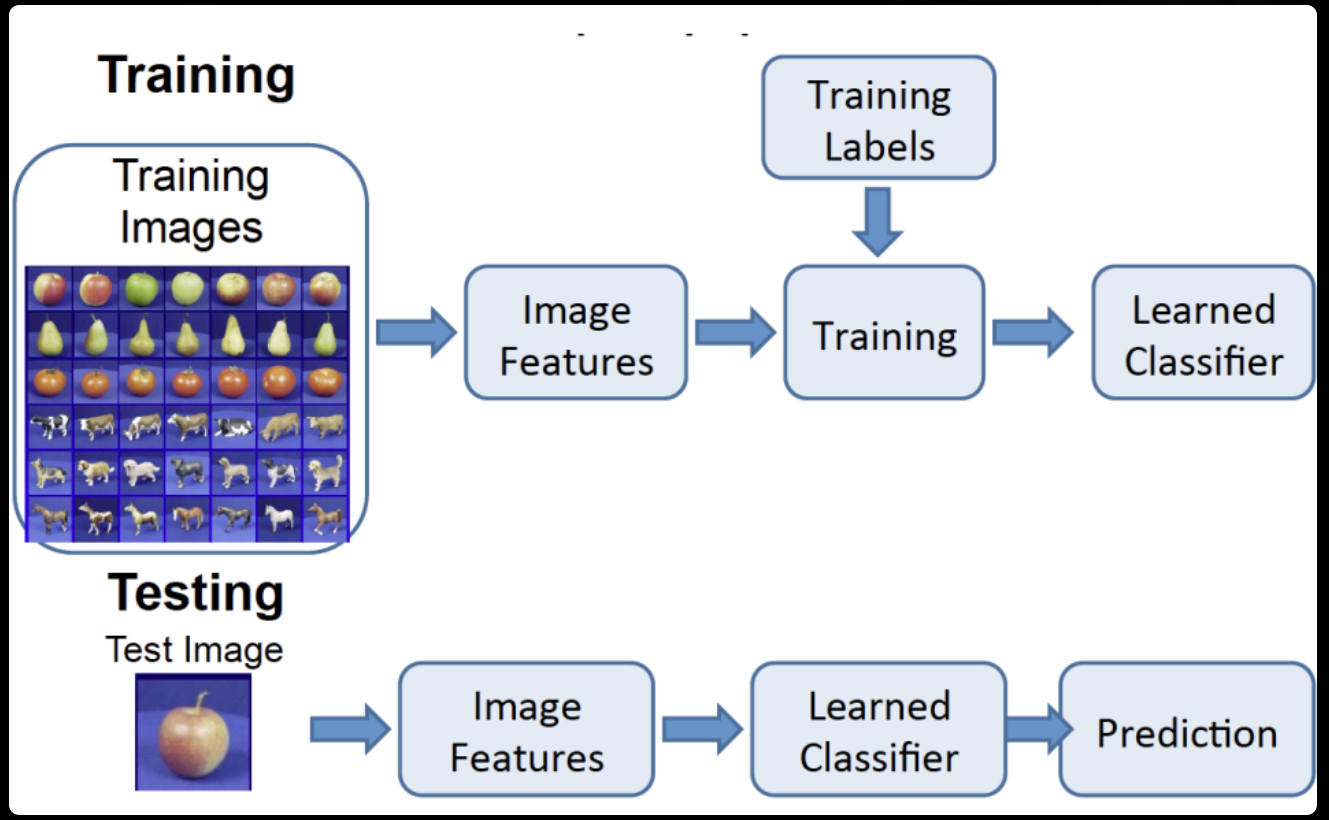


Image Classification Problem



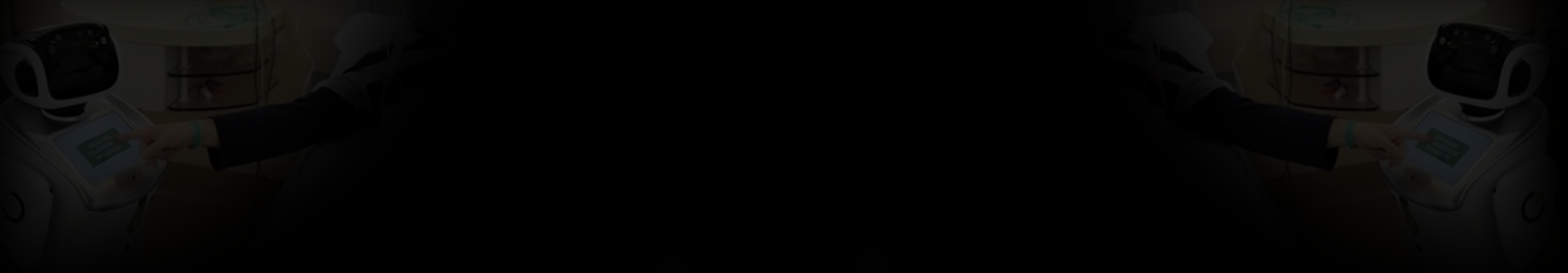
Machine Learning Solution

TUGAS

Dari dataset yang terdapat pada link berikut: **https://tinyurl.com/TugasComVis25**

Lakukan:

1. Ekstraksi ciri dengan hasil tabel, boleh menggunakan metode apa saja
2. Klasifikasi dengan menggunakan minimal 3 metode
3. Evaluation matrix terdiri dari akurasi, recall, precision dan f1-score



ANY QUESTION ?